



DSSG FINAL - RURAL E-MENTORING

Understanding Online Mentoring Relationships

Using NLP Techniques for Analysis and
Program Improvement

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THE TEAM



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Tiffany Chu



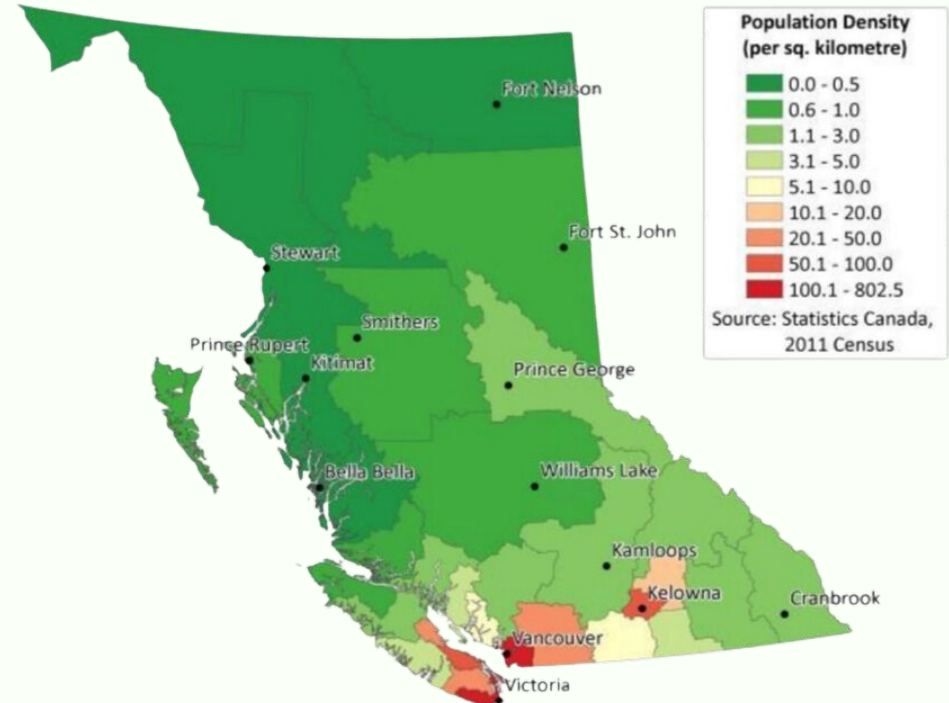
Jonah Curl



OUR PARTNER

RURAL EMENTORING BC: MOTIVATIONS

- Rural and Indigenous Communities have **higher demand for healthcare** but reduced access.
- Rural and Indigenous Students are more **likely to find careers in their communities**.
- UBC has programs designed to improve Rural and Indigenous admissions into Health Science



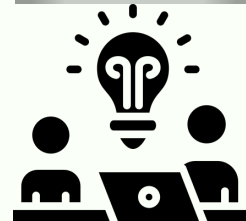
RURAL EMENTORING BC: MISSION AND

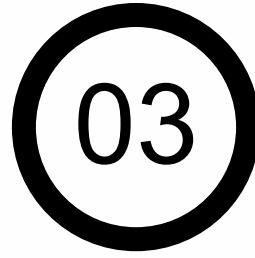


ReMBC's goal is to support and inspire new rural and indigenous healthcare practitioners in British Columbia.



- **Mentorship relationships:** one-on-one relationships through MentorCity's online mentoring (eMentoring) platform
- **Mentors:** university students in professional programs trained to support mentees through their career exploration
- **High School Curriculum:** to stimulate relationship development and conversation in various topics





PROJECT INTRO & GOALS

PROJECT INTRO & GOALS

Main Goal/Question :

**What are successful pairs doing differently that ReMBC
could help facilitate in the future?**

DATASET

- Access to all text conversations in eMentoring pairs via MentorCity
- Discussions within units/categories of the curriculum including:
 - Rural to Urban
 - Career Exploration
 - “Adulting”

Response Datetime	Relationship ID	Mentor	Response	Category
2024-01-01 10:03	1584937	Mentor	Hi, Hope you are well	Posts in Ways of Knowing
2024-01-01 12:00	1584937	Mentee	I am well, thank you	Posts in Ways of Knowing

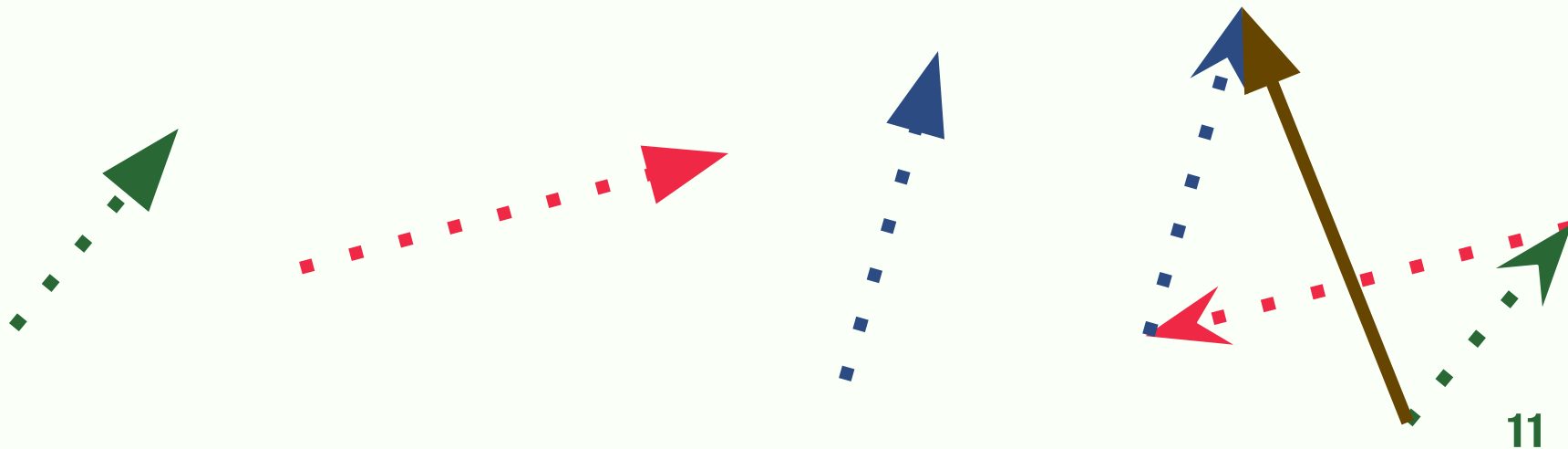
04

PROCESS & RESULTS

WORD EMBEDDINGS

Word Embeddings: words converted to multidimensional vectors, allowing for a measure of similarity using distance or angles.

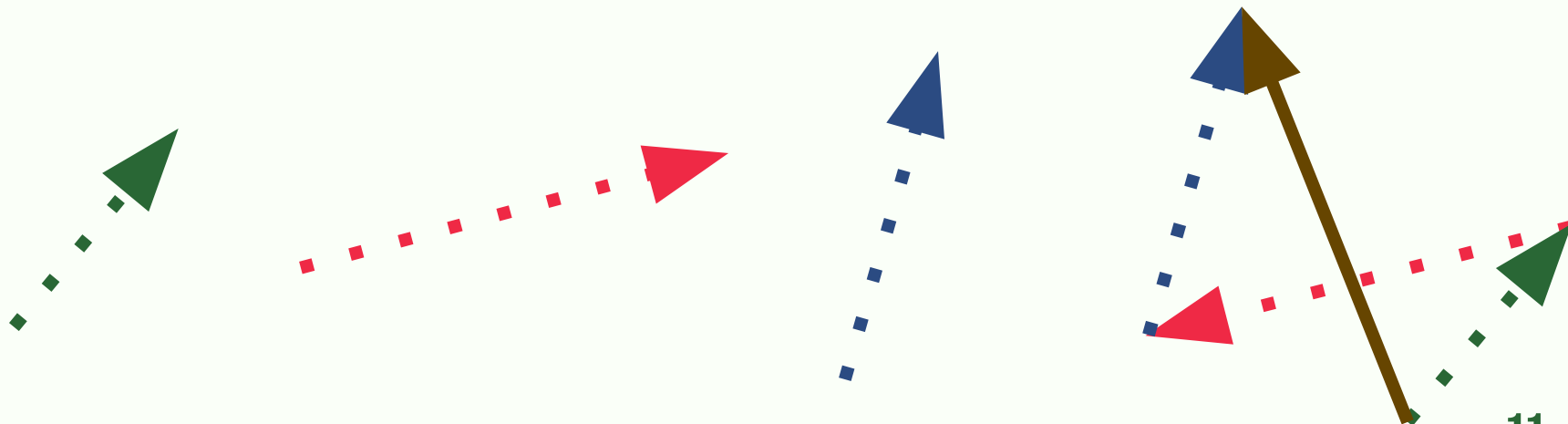
Kitten - Cat + Dog = ?



WORD EMBEDDINGS

Word Embeddings: words converted to multidimensional vectors, allowing for a measure of similarity using distance or angles.

Kitten - Cat + Dog = Puppy



1. TOPIC MODELING

Extracting common themes in conversations between mentees and mentors



Modular

- More customizable to specific needs

Bidirectional Embeddings (BERT)

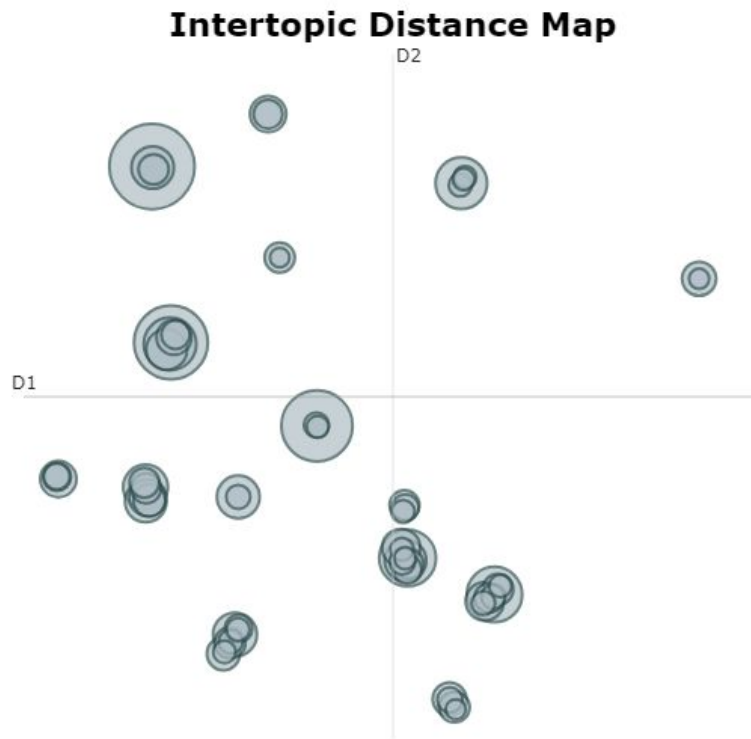
- Words are encoded based on both meaning and context in the document.

I am feeling blue. vs The water is blue.

blue $\neq$ blue

1. TOPIC MODELING: RESULTS

- Over 60 topics from mentee responses.
- Later topics have less responses in them.



1. TOPIC MODELING: RESULTS

BERTopic

Top 10 Topics From Mentee Responses:

1. **Career and Post-Secondary Exploration**
2. **Rural to Urban Living**
3. **Wellness and Self-care**
4. **Developing Good Study Habits**
5. **Mentorship Conversations**
6. **Free Time**
7. **Finding Inspiration**
8. **Hobbies**
9. **Online Platform**
10. **Career Exploration**

Representative Words:

careers, college, university
rural, towns, community
health, selfcare, stress
studying, memorize, practice
mentoring, talk, helping
spring, holidays, break
inspiration, passion, creation
hobbies, sports, skiing
login, messages, app
careers, nursing, profession

1. TOPIC MODELING: RESULTS

BERTopic

Top 10 Topics From Mentee Responses:

1. **Career and Post-Secondary Exploration**
2. **Rural to Urban Living**
3. **Wellness and Self-care**
4. **Developing Good Study Habits**
5. Mentorship Conversations
6. Free Time
7. **Finding Inspiration**
8. Hobbies
9. Online Platforms
10. Career Exploration

Representative Words:

careers, college, university
rural, towns, community
health, selfcare, stress
studying, memorize, practice
inspiration, passion, creation
careers, pursuing, profession

1. TOPIC MODELING: RESULTS



Top 10 Topics From Mentee Responses:

Representative Words:

1. Career and Post-Secondary Exploration

2. Rural to Urban Living

3. Wellness and Self-care

4. Developing Good Study Habits

5. **Mentorship Conversations**

mentoring, talk, helping

6. Free Time

7. Finding Inspiration

8. Hobbies

9. **Online Platform**

login, messages, app

1. TOPIC MODELING: RESULTS



Top 10 Topics From Mentee Responses:

Representative Words:

- | | |
|--|-------------------------|
| 1. Career and Post-Secondary Exploration | |
| 2. Rural to Urban Living | |
| 3. Wellness and Self-care | |
| 4. Developing Good Study Habits | |
| 5. Mentorship Conversations | |
| 6. Free Time | spring, holidays, break |
| 7. Finding Inspiration | |
| 8. Hobbies | hobbies, sports, skiing |

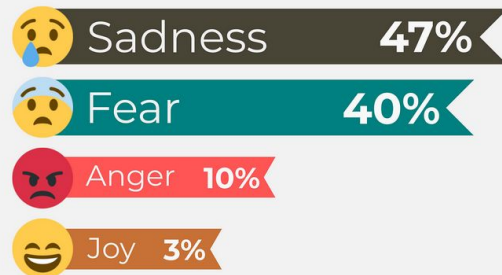
2. EMOTION DETECTION : HOW DOES IT WORK?

Method:

ZERO-SHOT
CLASSIFICATION



+ BART

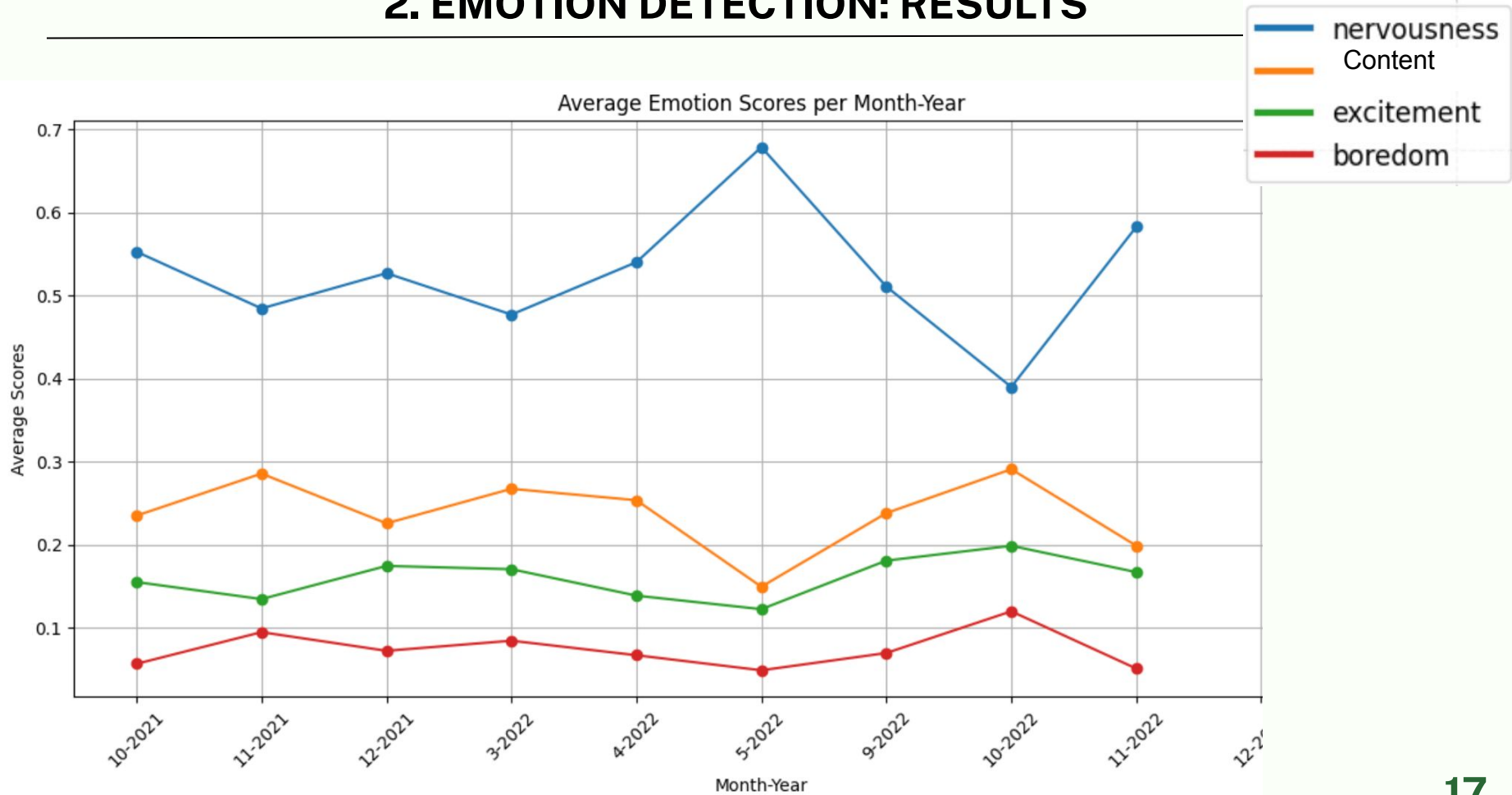


BART = **B**idirectional and **A**uto-**R**egressive

2. EMOTION DETECTION: RESULTS

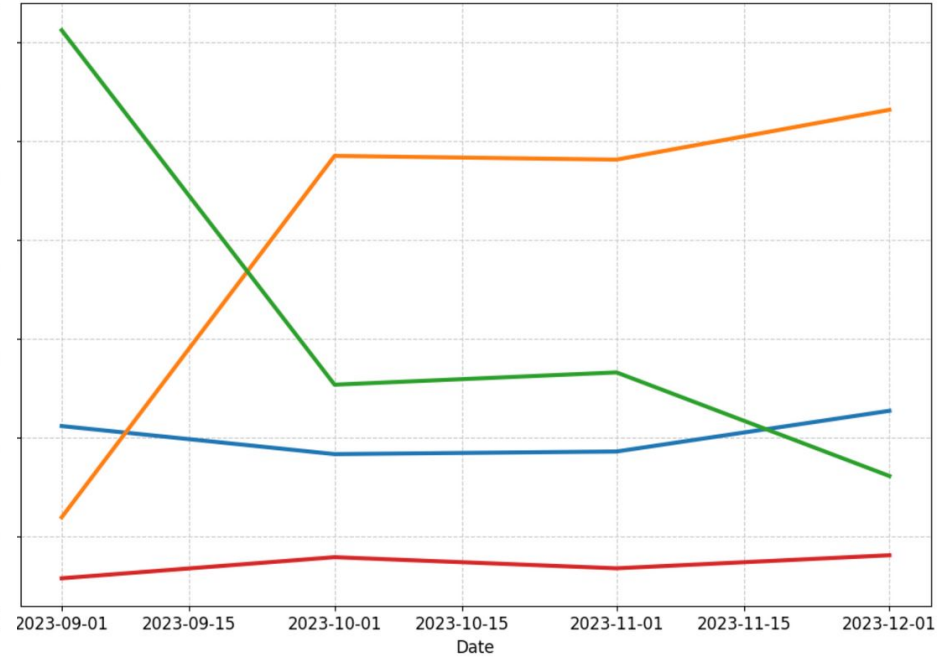
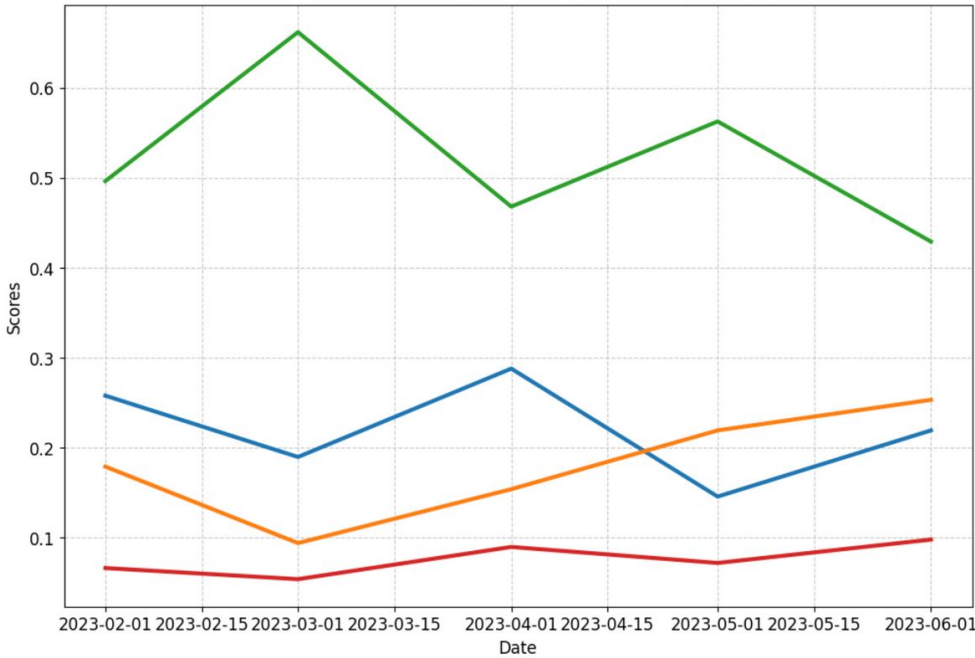
Datetime	Response	Predicted Labels	Scores
2021-09-14 16:41:00	Hey Mentor, I'm Mentee, bioengineering, wow. What do you enjoy about outdoor activities? My dad got me into them too, but skiing is definitely my favourite. I'm looking forward to getting to know you. I'm thinking we start with career considerations.	[excited, happy]	[0.342, 0.297]
2021-09-14 16:48:00	hi Mentor I'm a little shy and have a hard time and writing down what I am thinking. I'd like to start our conversation with choosing a program and school.	[nervous]	[0.857]

2. EMOTION DETECTION: RESULTS



2. EMOTION DETECTION: RESULTS

Mentee 1 vs Mentee 2's Emotions over time

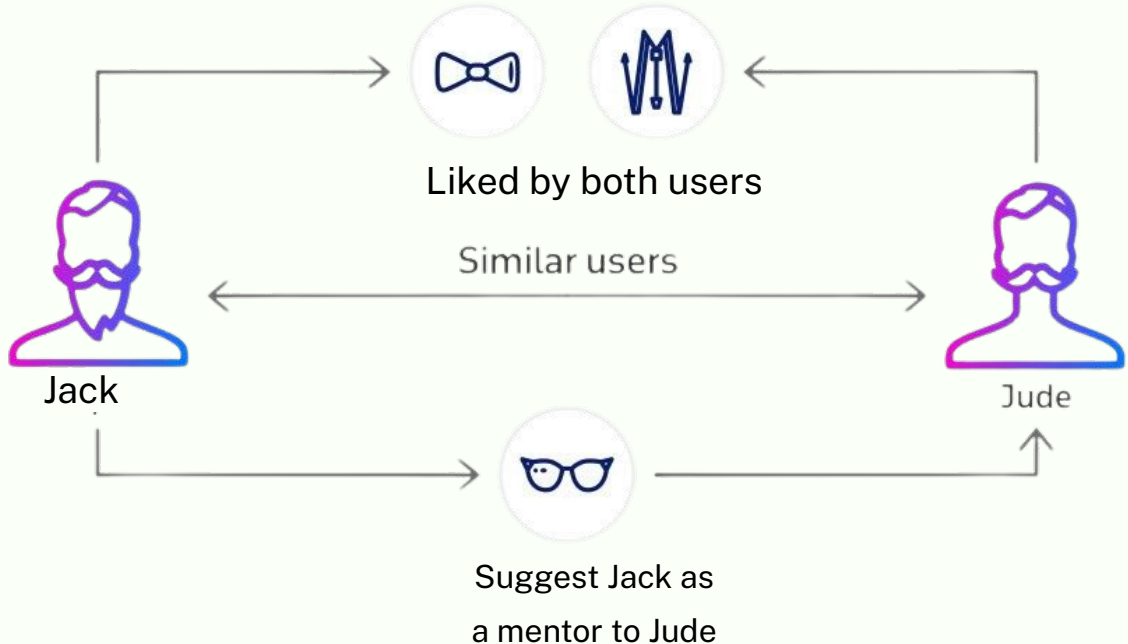


3.MENTOR-MENTEE MATCHING

Current process

Desired process

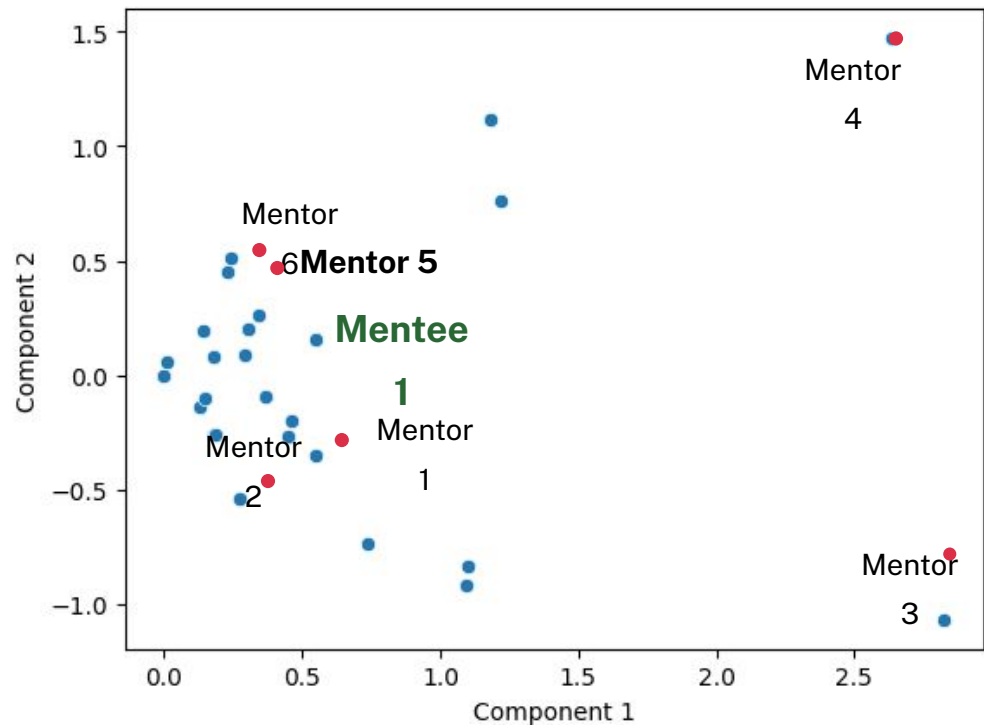
- Matching based on profile surveys
- Intensive Manual work



3.MENTOR-MENTEE MATCHING RESULTS

Person	Role	Nearest Neighbours
Bob	Mentor	[(Mary, 0.9, mentor), (Jane, 0.8, mentee) ...]
Jane	Mentee	[(Jack, 0.8, mentee), (Bob, 0.8, mentor) ...]

Output



Vector

4. ABSTRACTIVE SUMMARIES

Abstractive Summarization with Pegasus

Enter a paragraph of text to get its abstractive summary.

Input Paragraph

For as long as mankind has kept a history of its records, one can see humans have always yearned to explore. Whether it was the Nordic Vikings setting off to explore the seas, Christopher Columbus sailing for new lands, or Lewis and Clark blazing trails to the west, mankind has continually tried to push its boundaries and expand its frontiers. It would seem the call of adventure and discovery is something ingrained in human nature itself. Yet, as of late, this nature of exploration has been suppressed. Ever since the final frontiers of our earthly exploration were reached, and our maps of the "unknown" were finally filled in, man's pushing of boundaries has seemed to become stagnant. In this day and age, the new frontier for exploration lies in the next level of the unknown: space. Today's "Christopher Columbus" is Yuri

Summarize

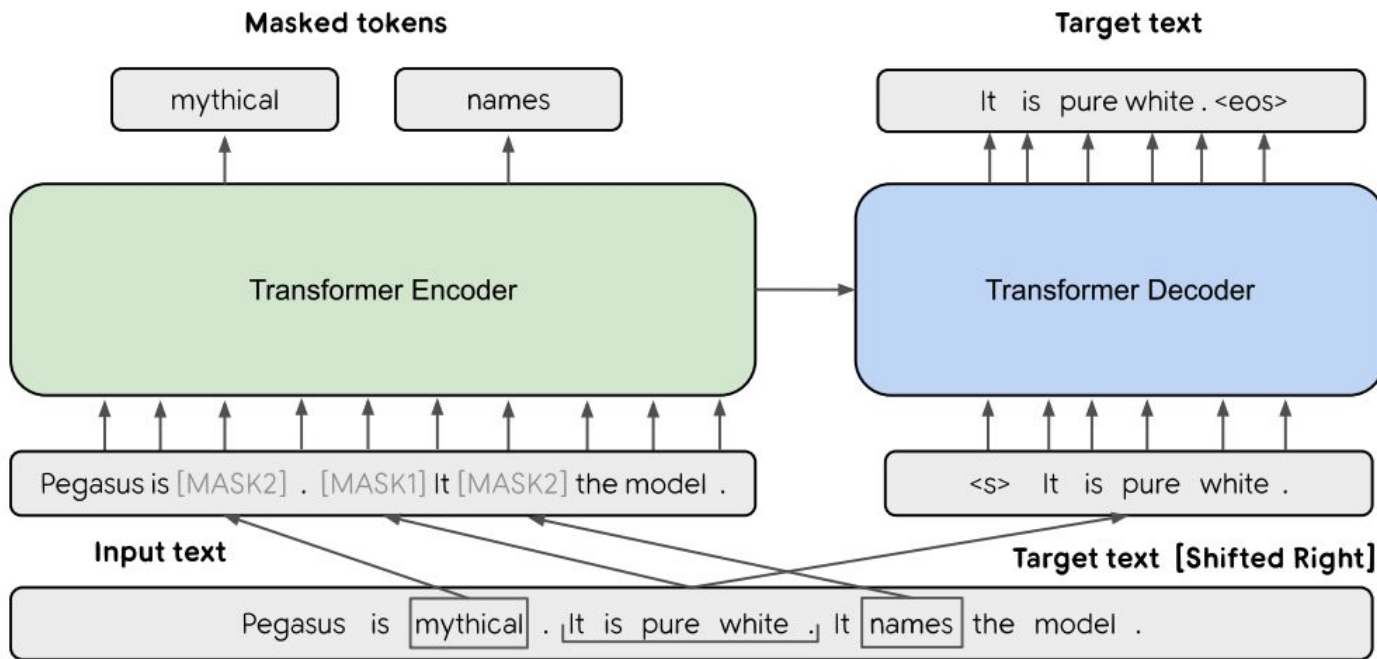
example input paragraph

Summary:

– The US has a long and proud history of space exploration, from Christopher Columbus and Lewis and Clark to Neil Armstrong and Yuri Gagarin. But in the past few years, the funding for space programs has been cut dramatically, leaving the US behind in the race to the moon and beyond, the Daily Beast reports. "Is this the way man's exploration and innovation should be treated? Certainly not," writes Andrew Sullivan in the Daily Beast. "Therefore, despite the protests of the opposition, funding for space programs must not be cut, as this would hinder mankind's advancement, survival, and exploratory nature."

output from our app

4. ABSTRACTIVE SUMMARIES



Pegasus Transformer showing Gap Sentences Generation (GSG)

5.DEPLOYING A NLP APPLICATION

This app allows researchers to upload their own files to use NLP on their own data without doing any of the coding themselves

Select the text column for context data

Response

Question Answering

Enter your question here

where do mentors go to school?

Answer

Answer: UVic

Topic 0

Topic 0 Topic 4 Topic 8 Topic 12 Topic 16

Enter a word to find related topics:

studying

Topics related to 'studying':

	Topic	Probability
0	0	0.6138
1	4	0.5275
2	8	0.4636
3	9	0.438
4	17	0.4094

query

NLP Preprocessing and Topic Modeling App

Upload a CSV file, select a text column for NLP preprocessing, and perform topic modeling to visualize the topics using BERTopic.

Upload a CSV file

Drag and drop file here
Limit 200MB per file • CSV

Browse files

csv upload

EMOTION DETECTION DEMO

Emotion Detection App

localhost:8501/mood2

Deploy

main

analysis

cleaning

codellamatest

mood

mood2

qa2

summarization

summarization2

topic modelling BERT

topic modelling LDA

Uploaded CSV file:

	Mentor ID	Mentee ID	Mentor Created at	Relationship ID	Response Datetime	Response
0	1,047,513,401	1,047,538,826	2020-09-28 09:32:00	40,140	2021-11-03 16:58:00	Hi Kendra! I fe
1	1,047,498,662	1,047,538,890	2019-11-01 17:04:00	40,144	2021-11-05 09:19:00	Hey teanna :)
2	1,047,516,499	1,047,540,775	2020-11-27 16:54:00	40,437	2022-03-09 13:49:00	Hi Rachel, Thi
3	1,047,541,741	1,047,541,040	2021-09-21 15:16:00	42,213	2021-12-16 11:26:00	Hey Brenna, I
4	1,047,549,093	1,047,548,897	2021-10-19 13:57:00	45,444	2022-01-10 14:01:00	Dear Japnaan

Select the text column for emotion detection

Response

Select a range of emotions to predict (3 to 9)

Choose an option

Select viewing format

☒ View each mentee's mood individually

☐ View all mentees' moods collectively (select 3 moods)

Perform Emotion Detection

ANALYSIS AND FILTERING DEMO

The screenshot shows a web browser window with the address bar displaying 'nlp-tool.streamlit.app/analysis'. The page features a sidebar on the left with a list of menu items: 'main', 'analysis' (highlighted), 'cleaning', 'mood', 'mood2', 'qa2', 'summarization', 'summarization2', 'topic modelling BERT', and 'topic modelling LDA'. The main content area is titled 'EDA (Exploratory Data Analysis)' and includes a sub-header 'Choose a CSV file'. Below this is a large light gray box with a cloud icon and the text 'Drag and drop file here' and 'Limit 200MB per file • CSV'. A 'Browse files' button is located to the right of this box. In the top right corner of the application, there are icons for 'Share', a star, a refresh icon, and a red status icon. At the bottom right, a dark button labeled '< Manage app' is visible.

Natural Language Processing

nlp-tool.streamlit.app/analysis

Share ☆ ↻ 🔴

main
analysis
cleaning
mood
mood2
qa2
summarization
summarization2
topic modelling BERT
topic modelling LDA

EDA (Exploratory Data Analysis)

Choose a CSV file

Drag and drop file here
Limit 200MB per file • CSV

Browse files

< Manage app

BERT TOPIC MODELLING DEMO

Natural Language Processing

nlp-tool.streamlit.app/topic_modelling_BERT

Share ☆ ↺ ⚠

Limit zoom per file • CSV

Master_2021-2023_Cleaned.csv 2.4MB

Update Dataset

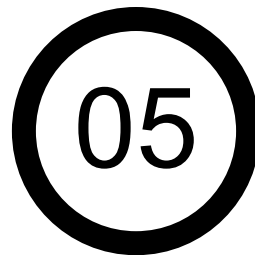
Uploaded CSV file:

	Mentor ID	Mentee ID	Mentor Created at	Relationship ID	Response Datetime	Response
22	1,047,516,499	1,047,540,775	2020-11-27 16:54:00	40,437	2022-03-30 21:41:00	Hello, I really
23	1,047,541,741	1,047,541,040	2021-09-21 15:16:00	42,213	2021-12-17 10:50:00	Hi Ava, I hope
24	1,047,549,093	1,047,548,897	2021-10-19 13:57:00	45,444	2022-01-10 22:26:00	Hi Ava, It is sc
25	1,047,513,321	1,047,549,007	2020-09-25 18:10:00	45,445	2022-01-13 08:21:00	Hi Avery, It's
26	1,047,547,979	1,047,548,888	2021-10-11 02:15:00	45,446	2022-01-12 16:53:00	Hi Dakota! Th
27	1,047,548,863	1,047,549,009	2021-10-17 09:37:00	45,449	2022-01-13 22:46:00	Hello Kady,
28	1,047,514,332	1,047,548,896	2020-10-14 16:46:00	45,450	2022-01-13 08:13:00	Do you know
29	1,047,547,951	1,047,548,884	2021-10-09 13:57:00	45,453	2022-01-12 12:59:00	I do agree wi
30	1,047,499,271	1,047,548,878	2019-11-11 16:28:00	45,455	2022-01-27 08:09:00	Dear Aziz, I h
31	1,047,548,854	1,047,549,414	2021-10-16 14:52:00	45,456	2022-01-10 19:13:00	Hey Shianne,

Select the column for topic modeling

Response

< Manage app



RECOMMENDATIONS

&

FUTURE DIRECTIONS

RECOMMENDATIONS & FUTURE DIRECTIONS



Label relationships
successful or unsuccessful.
Can introduce 3+ labels
(e.g. most engaged,
engaged, not engaged)

LABEL DATA



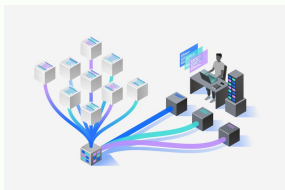
Apply tone detection to
mentor and measure
student engagement
accordingly. How do
different tones and writing
styles affect engagement ?

TONE DETECTION

05

CONCLUSION

CONCLUSION



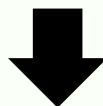
topic modeling



summarization



mood detection



university
guidance



curriculum
improvement
& training

ANY QUESTIONS?



THANKS!

